ANKU RANI

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EDUCATION

Massachusetts Institute of Technology, Cambridge, USA — September 2024 - Present
PhD in AI (GPA: 4.9/5.0)
Courses: 6.S975 Generative AI with Stochastic Differential Equations,
6.8300 Advances in Computer Vision, 6.8610 Quantitative NLP, MAS.S61 AI Visions,
MAS. S60 How to AI Almost Anything, MAS.630 Affective Computing.
Advisor: Dr. Paul Pu Liang, Co-Advisor: Dr. Pattie Maes

Plaksha University, Mohali, India — 2020 — 2021
Post Graduate Program, Artificial Intelligence
Advisor: Dr. Monojit Choudhury, Microsoft Research

Shivaji University, Maharashtra, India — 2015 — 2019
Bachelor of Technology, Electronics and Tele Communications
Minor in Data and Business Analytics

Research Experience

Massachusetts Institute of Technology — Cambridge, MA
Graduate Research Assistant — September 2024 — Present

- Research Project I: Culturally Yours [Demo]
  - Evaluated state-of-the-art video generation models for cultural faithfulness [Under Review COLM'26].
  - Pioneered culture transfer, built a framework to generate talking head video in different cultures using images and speech.
- Research Project II: Can Dialogues with AI trained with Persuasiveness improve humans' accuracy for discerning fake news? [Pilot Study]
  - Using HCI & dialogue systems to study how AI systems that uses persuasiveness affects humans' ability to distinguish between real and fake news [Best Paper Award, **CHI 2026**, **IC2S2, 2025**].

Adobe Research — Bengaluru, India
PhD Research Intern — Mentor: B Srinivasan (Adobe Research) — June 2024 — August 2024

- Research Project: Decoding Data Visuals: AI-Driven Attribution for Charts
  - Investigated Multi-modal Large Language Models' (MLLMs) capabilities in chart-based mathematical reasoning and introduced an attribution mechanism to enhance result interpretability.
  - This work spans Visual Question Answering (VQA), Visual Question Reasoning (VQR), and Visual Question Answer Attribution (VQAA) [Patent under Review, Paper accepted at **EACL'26**].

AI Institute, University of South Carolina — Columbia, USA
Research Specialist — Advisor: Dr. Amit Sheth — September 2023 - June 2024
Research Affiliate — December 2022 - September 2023

- Research Project I: Factify 5WQA — Enhancing Fact Verification with Aspect-Based 5W Explainability
  - Developed an aspect-based, human-interpretable, and explainable fact-checking system that employs a 5W (who, what, when, where, and why) question-answer framework [**Accepted at ACL 2023 Main Conference**]
  - Contributed to FACTIFY 3M dataset with a focus on 5W question-answering & image-text explainability addressing the need for multimodal fact verification system [**Accepted at EMNLP 2023 Main Conference**]
- Research Project II: SEPSIS: A new paradigm for deception detection
  - Investigated deception, particularly *lies of omission*, through a comprehensive framework that examines its types, colors, intentions, and topics
  - Developed a novel NLP-based deception detection model & explored the relationship between lies of omission and propaganda techniques to combat fake news [**Accepted at ACL SRW 2025**]
- Research Project III: AI-Generated Text detection for Indian Language
  - Addressing the concern of AI-generated text in regional languages, particularly Hindi by introducing a benchmark and dataset to assess the robustness of existing detection techniques [**Accepted at EMNLP 2024**]

NUS, Singapore — Remote
Research Assistant (Part-time)
Advisors: Dr. Kokil Jaidka, Dr. Sharath Chandra Guntuku — September 2023 — September 2024
These are the projects in collaboration with WWBP (World Well Being Project, University of Pennsylvania)

- Research Project I: Scientific Progress Influence on Public Discourse and Emotions: COVID-19 Vaccine Event and Twitter Discussion in the U.S.
  - Studied U.S. COVID-19 vaccine concerns post-first shots on a county level, examining public emotions through topic modeling, emotion analysis, and time series analysis on 7 months of Twitter data [Accepted at **JAMA Network Open, World Association for Public Opinion Research, IC2S2 2024**].
  - Analyze if emotions (e.g. anger, anxiety, incivility, etc) in vaccine tweets can predict vaccine hesitancy or vaccination rates at the county level
- Research Project II: Influence of the Twitter Acquisition Announcement by Elon Musk at the end of October 2022
  - Studying cross-cultural variations in mental health self-disclosures (MHSD) using TwitterDay data to understand global mental health communication

SELECTED PUBLICATIONS

C = Conference, J = Journal, P = Preprint/Submitted, W = Workshop — Google Scholar Link

[P.3] Durably Reducing Belief in Women's Health Misinformation Through Culturally Adaptive AI Videos
A Rani, K Jaidka, S Sharma, P Mahajan, M Wadhwa, A Lippman, P Maes, P Liang
Under Review CHI 2027 [Paper]

[C.9] CultureScore: Evaluating Cultural Faithfulness in Video Generation Models
A Rani, W Dai, S Nayak, P Maes, M Kalayeh, PP Liang
EMNLP 2026 [Paper]

[C.8] Dialogues with AI Reduce Beliefs in Misinformation but Build No Lasting Discernment Skills
A Rani, V Danry, PP Liang, A Lippman, P Maes
Best Paper Honorable Mention Award [Paper]

[J.1] **Collective Reactions to the Vaccine Rollout: Geolocating Emotional Shifts on Social Media**
K Jaidka, Y Wu, **A Rani**, O Kuru, R Jamalova, G Sherman, S C Guntuku
Accepted at Jama Network Open [Paper]

[C.7] **RADAR: A Reasoning-Guided Attribution Framework for Explainable Visual Data Analysis**
A Rani, A Garimella, A Saxena, B Srinivasan, PP Liang
EACL Findings 2026 [Paper]

[C.6] **SEPSIS: I Can Catch Your Lies – A New Paradigm for Deception Detection**
A Rani, D Dalal, S Gautam, P Gupta, V Jain, A Chadha, A Sheth, A Das
ACL SRW 2025 [Paper]

[C.5] **Can dialogues with AI systems help humans better discern visual misinformation?**
A Rani, V Danry, A Lippman, P Maes
IC2S2'25 [Paper]

[C.4] **Counter Turing Test (CT^2): Investigating AI-Generated Text Detection for Hindi - Ranking LLMs based on Hindi AI Detectability Index (ADI_{hi})**
I Kavathekar, **A Rani**, A Chamoli, P Kumaraguru, A Sheth, A Das
EMNLP Findings 2024 [Paper]

[C.3] **FACTIFY-5WQA: 5W Aspect-based Fact Verification through Question Answering**
A Rani, S.M Tonmoy, D Dalal, S Gautam, M Chakraborty, A Chadha, A Sheth, A Das
ACL 2023 Main Conference [Talk] [Poster] [Slide] [Paper]

[C.2] **FACTIFY3M: A Benchmark for Multimodal Fact Verification with Explainability through 5W Question-Answering**
M Chakraborty, K Pahwa, **A Rani**, S Chatterjee, D Dalal, H Dave, R Garimella, P Gurumurthy, A Mahor, S Mukherjee, A Pakala, I Paul, J Reddy, A Sarkar, K Sensharma, A Chadha, A Sheth, A Das
EMNLP 2023 Main Conference [Paper]

[C.1] **Use of Formal Ethical Reviews in NLP Literature: Historical Trends & Current Practices**
S Santy, **A Rani**, M Choudhury
ACL IJCNLP FINDINGS 2021 [Talk] [Poster] [Slide] [Paper]

[P.2] **A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models**
SM Tonmoy, SM Zaman, V Jain, **A Rani**, V Rawte, A Chadha, A Das
Under Review ACM Computing Surveys [Paper]

[P.1] **Visual Hallucination: Definition, Quantification and Prescriptive Remediations**
A Rani, V Rawte, H Sharma, N Anand, K Rajbangshi, A Sheth, A Das
Arxiv 2024 [Paper]

[W.1] **Memotion 3: Dataset on Sentiment and Emotion Analysis of Codemixed Hindi-English Memes**
S Mishra, S Suryavardan, P Patwa, M Chakraborty, **A Rani**, A Chadha, A Reganti, A Das, A Sheth, M Chinnakotla, A Ekbal, & S Kumar
Defactify Workshop at AAAI 2023 [Paper]

[W.2] **Factify 2: A multimodal Fake News and Satire News Dataset**
S Suryavardan, S Mishra, P Patwa, M Chakraborty, **A Rani**, A Chadha, A Reganti, A Das, A Sheth, M Chinnakotla, A Ekbal, & S Kumar
Defactify Workshop at AAAI 2023 [Paper]

Teaching Experience

Plaksha University Mohali, India
Teaching Assistant April 2022 — July 2022

- Projects and Applications for the Data Science 2 course (Instructor: Alexander Fred-Ojala, UC Berkeley)

Professional Achievements and Awards

- Received an Honorable mention Best Paper Award at CHI, 2026.
- Awarded a Tata fellowship (130k USD) for my research on the Arogya Vani Project, 2025
- Received a travel grant to deliver a parallel talk in Sweden at IC2S2, 2025
- Received AAAI 2024 scholarship to present in the convention center, Vancouver at AAAI 2024, and D&I scholarship to present at the Westin Harbour, Toronto at ACL 2023

Industry Experience

IBM Research Boston, USA
Visiting Student September 2026 — Present
Research Scientist Intern June 2026 — August 2026

- Training Multimodal LLMs for interactive character-based communication
- Trained LLMs for Quantum Circuit Synthesis

Shaip AI DATA LLP Ahmedabad, India
Consultant August 2023 — September 2023
Research Scientist November 2022 — July 2023

- Worked on n-gram and RPE architecture for clinical data with improved long-term dependency handling

Cactus Labs, Cactus Communications Pvt Ltd Mumbai, India
Software Engineer(NLP) May 2022 — September 2022

- Developed & deployed a rule-based consistency module to production for manuscript assessment; accuracy 90%

Verisk Analytics Gurgaon, India
Data Scientist I September 2021 — April 2022

- Created text model framework with 92% accuracy using keyword matching for underwriting business
- Experimented on boolean question & answering model using SOTA to reduce manual efforts; accuracy 87%

Apptio, An IBM Company Bengaluru, India
Product & Engineering Intern April 2021 — June 2021

- Built React UI for AWS deployment, saving 2000+ developer hours annually and forecasted AWS resource utilization for 2 weeks, enabling efficient resource planning using time series modeling

PixiuAI LLP Bengaluru, India
Data Scientist Trainee June 2020 — November 2020

- Founding Member, launched MVP to find finance news for stock value insights
- Constructed news feed with 87% precision, boosting user retention by 22%

Civilsdaily New Delhi, India
Data Analyst January 2019 — May 2020

- Leveraged text modeling techniques (text similarity, custom entity recognition) for the UPSC ed-tech startup with 100k+ online visitors (MAUs) & 50k+ paid users

NITI Aayog, Government of India Singrauli, Madhya Pradesh
District Collector Intern June 2018 — August 2018

- Built and deployed framework to monitor health and education in the district; reported 29% YoY increment in institutional deliveries

Academic Service

- Reviewer (CHI'27, CHI'26, COLM'26, ACL ARR 2026, ACL ARR 2025, IC2S2, NAACL'24, ACL'24, IC2S2'24), program committee member for ACL'23 & ACL'24 clinical NLP workshop
- Co-organizer of the Defactify workshop (multimodal fact-checking and hate speech detection) at AAAI' 23 & AAAI'24
- Volunteer at AAAI'24, ACL'23, EACL'23, AAAI'23, ACL'21, EMNLP'21, Emoji'21 ICWSM
- Mentored 10+ women in Plaksha TLP'22, boosting their average salary offers by 20%

Media Coverage

Invited Talks

KU Leuven
Durably Reducing Belief in Women's Health Misinformation Through Culturally Adaptive AI Videos. October 2026

Boston AI, ML and Computer Vision Meetup
Beyond the Black Box: Vision Language Models that Explain & Empower February 2026

EmTech MIT 2025
Localization of Videos using Generative AI October 2025

The 11th International Conference on Computational Social Science (IC2S2), Sweden
Talk Title: Can dialogues with AI systems help humans better discern visual misinformation? July 2025

SAP Sapphire Madrid
Talk Title: Mind & Machine: Imagining the future of human creativity with AI May 2025